



Contrastive Retrieval Methodology for Turkish Metaphor Detection and Identification

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Metaphorical expressions, as a form of figurative language, are individually limited in their use. However, when both literal and non-literal meanings are considered, they are frequently used in web content. Hence, producing a balanced dataset to learn superior representations is a challenging task, and metaphor detection suffers from a limited training dataset. To alleviate this problem, we present a retrieval-based contrastive learning approach which first identifies candidate metaphors in the input text and then detects metaphorical expressions as a claim verification task in the inherently unbalanced setting of this study. Furthermore, we adapt contrastive learning to make it easier to distinguish between the literal and figurative meanings of the same expression. For the experimental setup, we extract non-literal and literal expressions along with their meanings and sample sentences from a Turkish dictionary. In the metaphor detection subtask, performance evaluation shows that sparse and dense search variations using the Turkish-e5-Large model achieve a Recall@10 (R@10) score of 0.614. Moreover, the SimCSE-TR-Contr-Sample-Meaning model achieves the highest Recall@10 (R@10) of 0.9739 on the generated test dataset for the metaphor identification subtask. In the real-world scenario, it achieves a competitive R@10 score of 0.8684, and these results clearly demonstrate that our model can generalise to this real-world scenario.

CCS Concepts: • **Computing methodologies** → **Natural language processing**;

Additional Key Words and Phrases: Metaphor detection, Contrastive learning, Turkish metaphor dataset

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1 Introduction

Figurative language expressions require more complex processing than general natural language tasks, and the detection of these expressions often suffers from limited training datasets. Metaphorical terms, as one type of figurative language expression, are commonly used in everyday life. However, the inherently literal meanings of the target metaphorical concepts are widely available in web content, making it a difficult task to generate a balanced data resource for accurately training their representations [24]. To address this challenge, claim verification presents a promising direction for handling this unbalanced setting, as it retrieves a large set of textual sources to form evidence from web content [44].

To retrieve relevant claim-evidence pairs from a large dataset, open-domain passage retrieval, which uses a different data source by extracting question-passage pairs, has attracted attention

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in recent years. By employing the Bi-Encoder mechanism, representations of these pairs can be precomputed at scale, enabling retrieval with satisfactory accuracy [31]. **Dense Passage Retrieval (DPR)** proposes a contrastive learning methodology for training dense passage representations, aiming to maximise the dot product between two distinct BERT encoder models [28].

The main contributions of our work can be summarised as follows: First, we generate a Turkish metaphor detection dataset from the Turkish TDK dictionary to construct training and test sets. We extract terms from the Turkish WordNet [18] and match them with entries in the Turkish Wiktionary using an exact string matching algorithm. Additionally, we compute the Jaccard similarity between the definitions of the terms in WordNet and the Turkish Wiktionary to obtain a more robust term list. Finally, we employ contrastive learning to facilitate the distinction between the literal and figurative meanings of the same expression through a combination of sparse keyword-based search and dense passage representations.

The article is structured as follows: Section 2 presents related work, and Section 3 introduces the proposed retrieval-based contrastive learning method for metaphor detection. Section 4 outlines the evaluation of the dataset and presents experimental results. Section 5 discusses potential real-world applications of the metaphor detection task. Finally, Section 6 concludes the study and suggests future directions.

2 Related Work

Linguistic metaphor rules, such as the **Metaphor Identification Procedure (MIP)** [23] and **Selectional Preference Violation (SPV)** [34], are widely employed in both metaphor identification and detection tasks. The MIP theory focuses on differentiating the literal meaning of a target term from its contextual meaning, whereas the SPV theory identifies metaphorical usage by analysing irregularities in typical word co-occurrences.

MelBERT [11] introduces a metaphor detection model that frames the task as a supervised learning problem, leveraging the MIP and SPV theories. This model tackles the classification task of identifying whether a target word in a sentence is used metaphorically. Similarly, MisNet [55] enhances the representation of these two linguistic rules by treating them as separate semantic matching tasks in metaphor detection. It uses MIP as a representation-based semantic matching task, comparing the target word in the sentence to its basic usage across two encoders. Moreover, it models SPV as an interaction-based semantic matching task, measuring the relationship between the target word and its context. Although these approaches have demonstrated promising results, a key limitation is that supervised learning-based metaphor detection methods struggle with generalisation, particularly in more complex contexts.

To overcome this limitation, our first contribution is the application of contrastive learning to distinguish between literal and metaphorical usage, as it has shown promising results in downstream tasks such as question answering and retrieval augmentation. As a key contribution to contrastive learning, SimCSE [21] employs dropout to augment data by processing the same sentence twice in an unsupervised learning mechanism, thereby facilitating the distinction between literal and metaphorical usage. In this mechanism, embedding pairs of the same sentence are treated as positive instances. Additionally, the **Stanford Natural Language Inference (SNLI)** [7] and **Multi-Genre NLI (MultiNLI)** [51] datasets provide semantically similar sentences that entail each other, serving as positive instances, while sentences with contradiction labels are treated as hard negative instances. Another contribution of our study is the use of a Turkish NLI model to rerank query results for candidate metaphors as part of a question answering mechanism. In terms of reranking methods for question answering, Chuang et al. [12] propose a query reranking method for open-domain neural question answering by combining GAR [35] and BM25 [40] sparse retriever models.

Additionally, Kratzwald et al. [30] introduce RankQA, a three-layer method for reranking answers in neural question answering.

Our retrieval-based contrastive learning method relies on robust text similarity techniques, with several promising approaches developed to measure short-text similarity. For instance, Sun et al. [42] propose a surrogate model trained on automatically labelled pairs of sentences, calculating the probabilities of two sentences based on the same context. For languages other than English, Tsukagoshi et al. [45] introduce Japanese SimCSE, which fine-tunes SimCSE for the unsupervised setting by leveraging datasets such as Wiki40B [22], Wikipedia, and CC100 [15]. It also uses a **Japanese translation of SNLI (JSNLI)**¹, JaNLI [54], and a ChatGPT-translated version of the SNLI and MultiNLI datasets for the supervised setting.

Several studies have investigated the use of figurative language in Turkish. A notable example is euphemism detection, in which more unpleasant or offensive terms are replaced with polite or indirect expressions. Euphemisms serve to make unpleasant concepts sound less severe, such as using ‘passed away’ instead of ‘died’. Biyik et al. [5] introduce a dataset of Turkish euphemistic terms and conduct experiments with Transformer-based models for euphemism detection in Turkish, leveraging their dataset. Another example is irony detection, where a sentiment is expressed by stating the opposite of the intended literal meaning. Bölücü and Can [6] explore the use of Graph Neural Networks for detecting irony in Turkish, utilising semantic information from the Universal Conceptual Cognitive Annotation. As demonstrated in our previous work [26], we also explored the application of graph representation learning for metaphor detection. In our initial study, we used the Node2Vec model to generate feature vector representations, which we then applied to supervised learning techniques, such as logistic regression for detecting metaphors in Turkish texts. Since this approach relies on supervised learning techniques, it requires a substantial amount of balanced training data. However, it is not well-suited for generating a balanced dataset, as the literal meanings of the target metaphorical concepts are widely available in web content, leading to an imbalance between metaphorical and non-metaphorical instances.

To the best of our knowledge, this is the first attempt to detect metaphors in Turkish texts using a retrieval-based contrastive learning method for metaphor detection. We employ this method to distinguish between the contextual and basic meanings of the target terms, generating a Turkish metaphor dataset for the contrastive learning step and eliminating the need for a training phase. We compare our method with MelBERT and MisNet, two state-of-the-art English models, using our generated Turkish metaphor dataset. Additionally, we compare it with English datasets such as MOH-X [36], which includes text samples where a verb is labelled as metaphorical or literal, extracted from WordNet, and TroFi [4], a verb metaphor detection task generated from the Wall Street Journal Corpus Release. Since MelBERT and MisNet are supervised learning-based models, they require balanced and well-defined training samples. In contrast, we employ contrastive learning to better distinguish between the literal and figurative meanings of the same target term in this unbalanced setting, treating it as a zero-shot task with only a few samples to adapt the contrastive learning.

3 The Proposed Method

Metaphor detection and identification are two main subtasks of our work; both aim to detect metaphorical usage in a given text. To distinguish the objectives of these subtasks, Gao et al. [20] define the metaphor identification task as the sequence labelling task, which involves predicting whether each word in a sequence is metaphorical or literal without any prior information about the target term. As the second subtask, the metaphor detection entails classifying whether a target term in a given sentence is metaphorical.

¹<https://nlp.ist.i.kyoto-u.ac.jp>

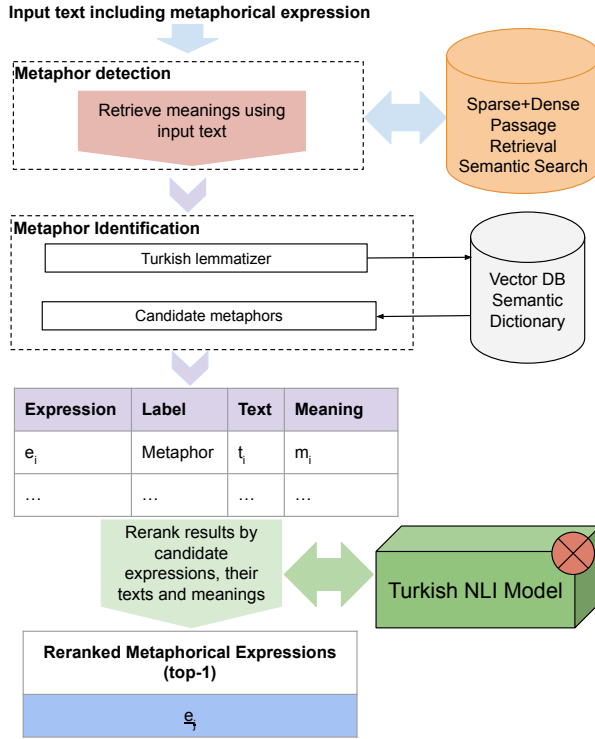


Fig. 1. The general architecture of the proposed method.

Figure 1 illustrates the overall structure of the proposed method, which is adapted from the fact-checking methodology [44] and the aspect-based document similarity method using a pairwise document classification task [37]. The latter uses a series of language models for research studies from the ACL Anthology and the CORD-19 corpus. Instead of performing a binary classification task to distinguish between similar and dissimilar citations, they utilise the section title, which includes a citation and a label for the given document pair. In contrast to using section title information, we retain the target terms with their textual representations, including meanings and sample texts in which these terms appear. Furthermore, we apply contrastive learning to differentiate between similar and dissimilar textual representations, reranking query results by leveraging a Turkish **Natural Language Inference (NLI)** model sourced from the NLI-TR dataset [8].

The main assumption of our work is that the claim or input text contains at least one metaphorical expression, which is then split into stemmed tokens using the Stanza library [38] for Turkish (see Table 3 for a detailed comparison). After, we use Chroma² as the vector database to build a metaphorical expression dictionary for finding candidate metaphors in the claim text. A vector database stores high-dimensional vector embeddings and is designed to handle unstructured data, where each entry points to a vector in a multidimensional space with a fixed number of dimensions. Unlike traditional relational databases that organise data into rows and columns, vector databases represent data points as vectors. In the candidate metaphor detection subtask, we search Chroma,

²<https://docs.trychroma.com/>

the open-source embedding database, acting as the Vector DB Semantic Dictionary, for terms that have metaphorical (but not literal) meanings.

Once the candidate metaphorical expressions are identified, we perform two parallel steps to detect similar metaphors: checking the semantic similarity either between the claim text and the expression meanings, or between the candidate and expression meanings in the dense passage retriever [28]. First, we employ the sparse BM25 model [40], which searches these expressions, followed by the DPR model, which semantically searches the corpus, including both metaphorical and literal meanings. The queries consist of these expressions, along with the claim text or their meanings. Thus, this step mainly re-ranks the meanings in the corpus by using the contrastive loss function, as described in Equation (1). The main objective of the metaphor detection step is to determine the meaning of the metaphorical expression in the claim text by adapting it to a typical relevance judgment methodology.

The main goal is for a sentence containing a potential non-literal expression to be semantically close to the metaphorical meaning and distant from the literal meaning in the vector space. As the training objective, our method uses the normalised temperature-scaled cross entropy [21].

$$L = -\log \frac{e^{\text{sim}(hs_i, hm_i^{(+)})/\tau}}{e^{\text{sim}(hs_i, hm_i^{(+)})/\tau} + \sum_{j=1}^N e^{\text{sim}(hs_i, hm_j^{(-)})/\tau}}, \quad (1)$$

where hs_i is a sentence containing the target term, $hm_i^{(+)}$ and $hm_i^{(-)}$ are the semantic representations of gold positive and hard negative meanings, respectively. To compute the similarity between sentence-meaning pairs, $\text{sim}(h_1, h_2)$ represents the cosine similarity and τ is a temperature hyperparameter. During sentence encoding, BERT-based transformer models [16] are used and fine-tuned using Equation (1) as the contrastive target. To measure the quality of contrastive learning, we apply Equation (2) for alignment and Equation (3) for uniformity. Equation (2) depicts the alignment loss L_{align} in terms of f and α .

$$L_{align}(f; \alpha) \triangleq -\mathbb{E}_{(hs_i, hm_i^{(+)}) \sim p_{pos}} [\|f(hs_i) - f(hm_i^{(+)})\|_2^\alpha], \quad (2)$$

where $\alpha > 0$ and the alignment loss refers to the expected distance between positive pairs [50]. Equation (3) shows the uniformity loss as the logarithm of the average pairwise **Radial Basis Function (RBF)** kernel.

$$L_{uniform}(f; t) \triangleq \log \mathbb{E}_{(u, v) \sim p_{data}^{(i.i.d)}} [G_t(u, v)], \quad (3)$$

where t is a fixed parameter, u and v refer to the embeddings of hs_i and hm_i . Finally, Equation (4) defines the RBF kernel. Hence, the distribution that optimises the uniformity metric converges to a uniform distribution, which is empirically reasonable with a finite number of data pairs [50].

$$G_t(u, v) \triangleq e^{2t \cdot u^T v - 2t}, \quad (4)$$

where $t > 0$, and G_t refers to the average pairwise Gaussian potential, which quantifies how spread out the embeddings are and is associated with the uniform distribution on the hypersphere within the latent space.

3.1 Candidate Metaphor Detection

We collect metaphorical expression-meaning pairs by measuring the similarity of the input text using a semantic search mechanism [43], which enables the combination of sparse [40] and DPR [28]. We index these pairs using the BEIR framework, which can be adapted for dense retrieval architectures

as the semantic search mechanism. From this framework, we utilise dense retrieval models, including BERT, ELECTRA [13], and LaBSE [19], as similarity-based pre-trained language models.

Our retrieval-based contrastive learning approach first identifies candidate metaphors in the given text and then detects metaphorical expressions as part of a claim verification task [44]. A typical claim verification task consists of two steps: claim generation and labelling. In the claim generation step, information is extracted from online knowledge sources such as Wikipedia. The second step involves annotating these extracted claims with appropriate labels, such as supported, refuted, or not enough information. To adapt this task to candidate metaphor detection, we extract information, including text samples and meanings of both literal and metaphorical expressions, from the Turkish dictionary. Instead of assigning labels for the claim verification task, we assign metaphorical or literal labels to the candidate expressions.

In this step, there is no prior information about which term in the input text refers to a metaphorical expression before the identification task. Furthermore, the retrieved query results may include both literal and metaphorical examples from the given corpus. As a result, it may retrieve query results for potential metaphorical expressions along with their meanings by comparing similarities between the input text and the meanings.

3.2 Candidate Metaphor Identification

When identifying candidate metaphors, we use the Stanza library [38] to split the input text, which may contain a potential metaphorical expression, into tokens. Since these input texts are collected from the Turkish dictionary, they primarily consist of one-sentence examples and short meanings. We then use the Turkish lemmatiser to transform each token into its stemmed form, making them more suitable for querying the Vector DB Semantic Dictionary, which is curated from metaphorical expressions along with their sample texts and meanings.

As an open-source vector embedding database, we use Chroma, which stores embedding models along with their metadata. This makes it suitable for filtering keywords based on their meaning-based embeddings. In this case, we create an embedding dictionary by loading all metaphorical (term, meaning) pairs into the Chroma database before searching for candidate metaphorical expressions.

To filter out candidate expressions (e.g., e_1 with sample text t_1 and short meaning m_1 in Figure 1), we match the candidate metaphors with the retrieved meanings from the candidate metaphor detection task. If no metaphorical expression is found in the input text after identifying candidate metaphors, our method automatically assigns the literal label to the input text. Otherwise, it reranks the query results for candidate metaphorical expressions and applies the Turkish NLI model to identify the most distinctive expressions using the contrastive learning method.

3.3 Candidate Metaphor Reranking

We adapt the NLI-TR dataset [8] for the reranking subtask by using the sample texts and meanings of the candidate expressions after the metaphor identification task. The NLI-TR dataset consists of Turkish translations of the SNLI [7] and MultiNLI [51] datasets, where the premise and hypothesis sentences correspond to the input text and either the sample text or the meaning of each candidate entity, respectively. Re-ranking the initial query results becomes a sentence pair prediction subtask, and we rerank the results based on their inference scores.

To collect the inference scores of the pre-trained language models, we further explore Turkish pre-trained models such as BERT [16] and DistilBERT [41], with the latter retaining the performance of the original model in a smaller version [1]. Additionally, ConvBERT [27] is another large language model trained on the Turkish portion of the larger multilingual C4 corpus [53]. For reranking the query results, we employ a similar mechanism [9] using the HuggingFace pipeline [52].

4 Experimental Setup

4.1 Dataset

To generate training data for fine-tuning NLI models and a test dataset for comparing our zero-shot method with other studies, we retrieve terms from the online Turkish dictionary.³ Since this dictionary relies on a search mechanism rather than providing a complete list of terms, we use the Jsoup library to collect a list of terms from the Turkish Wiktionary, which includes entries from A to Z. Once the term list is gathered, we query the Turkish dictionary for each term to obtain its meaning, a classification label as either literal or metaphorical, and a sentence containing the term (see Appendix A for a detailed, step-by-step description of how the dictionary was scraped and cleaned from an algorithmic perspective).

Table 1 presents the information extracted from the Turkish dictionary for the target term “kalp” (heart), along with its English translations. In Turkish, the literal meaning of “kalp” refers to the organ responsible for pumping blood throughout the body. Metaphorically, it can also represent emotion or feeling, as illustrated in the second example.

4.2 Evaluation Metrics

We use the metrics NDCG@k, MAP@k, and Recall@k, where k represents the top-k query results. The first metric, **Normalised Discounted Cumulative Gain (NDCG)**, evaluates ranking effectiveness by prioritising higher-ranked query results over lower-ranked ones. It takes into account the position of relevant items in the ranking, with values ranging from 0 to 1, where higher values indicate better performance.

$$NDCG@k = \frac{\sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}}{\sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}}, \quad (5)$$

where rel_i represents the i^{th} relevance value, which is either 0 or 1 in our experimental setup, with 0 indicating literal use and 1 indicating metaphorical use of the given text, respectively. The value of k in the numerator reflects the results in their actual order, while the k in the denominator represents the results in the expected order. MAP@k is a variation of **Average Precision (AP)**, calculated as the mean of the individual AP scores.

$$MAP@k = \frac{1}{N} \sum_{i=1}^N AP@k_i. \quad (6)$$

AP@k is then calculated as

$$AP@k = \frac{1}{\min(k, rr_{total})} \sum_{i=1}^k \frac{rr_{seen}}{i}. \quad (7)$$

Recall@k is an unranked metric and represents the fraction of retrieved relevant query results rr_i out of the total relevant results, which include both retrieved relevant results rr_i and non-retrieved relevant results $rnotr_i$ for the top results, as denoted in Equation (8).

$$Recall@k = \frac{rr_i}{rr_i + rnotr_i}. \quad (8)$$

As an adaptation of the information retrieval task, we transform the joint metaphor identification and detection tasks into a mechanism that delivers the most relevant retrieval results to domain experts as they evaluate the input text. We run all our experiments on BEIR, a heterogeneous

³<https://sozluk.gov.tr/>

Table 1. Extracted Information about the Target Term: “kalp (heart)”

Property	Lang.	Meaning	Sample text
Gerçek	Tr	Göğüs orta boşluğunda, iki akciğer arasında, vücudun her yanından gelen kirli kanı akciğerlere ve oradan gelen temiz kanı da vücuda dağıtan organ, yürek	Bak ellerim nasıl titriyor, bak alnım nasıl yanıyor, bak kalbime nasıl çarpıyor.
Literal	En	The heart is the organ in the middle of the chest, between the two lungs, that distributes the dirty blood coming from all over the body to the lungs and the clean blood coming from the lungs to the rest of the body.	See how my hands are shaking, see how my forehead is burning, see how my heart is pounding.
Mecaz	Tr	Duygu, his	İnsanı tekrar, kalp ve fikir cennetine erdirtirebilecek tek kudret kadındır.
Metaphor	En	Emotion, feeling	The only power that can bring man back to the heaven of heart and mind is woman.
Gerçek	Tr	Düzme, sahte, geçmez (para)	Aslı ile kalpı arasındaki farkı ancak o işin çilekeşleri bilir.
Literal	En	Counterfeit, fake, illicit (money)	Only those who suffer in this realm know the difference between the original and the heart.
Mecaz	Tr	İşe yaramaz, tembel	Kalp adam.
Metaphor	En	useless, lazy	Heart man.

benchmark that includes a variety of information retrieval tasks [43]. All benchmark results are obtained on both CPU (Intel i7-4770) and GPU (GeForce GTX 1070 8 GB) using Hugging Face Transformers.⁴ For all pre-trained transformers, we use base models.

4.3 Comparison of Pre-trained Language Models for Candidate Metaphor Detection

In this evaluation, we use the ConvBERTurk model provided by the MDZ Digital Library team (dbmdz) at the Bavarian State Library,⁵ a Turkish community-driven cased model. The Distiluse model⁶ encodes sentences into a 512-dimensional dense vector space and is well-suited for tasks such as clustering or semantic search. The Geotrend/bert model⁷ is a smaller version of the BERT base model, designed to produce representations identical to the original model while maintaining the same accuracy [1]. ELECTRA [13] improves on traditional pre-training methods by replacing

⁴<https://huggingface.co/transformers/>

⁵[dbmdz/convbert-base-turkish-cased](https://dbmdz.org/convbert-base-turkish-cased)

⁶<https://huggingface.co/sentence-transformers/distiluse-base-multilingual-cased-v1>

⁷<https://huggingface.co/Geotrend/bert-base-tr-cased>

Table 2. Comparison of Candidate Metaphor Detection Models

Model	NDCG@3	NDCG@10	M@3	M@10	R@3	R@10
ConvBERTurk-Dense	0.0142	0.0229	0.0029	0.0063	0.0174	0.0435
Distiluse-Dense	0.0567	0.0796	0.0594	0.0734	0.0783	0.1391
Geotrend/bert-Dense	0.0316	0.0496	0.0116	0.0168	0.0316	0.0956
ELECTRA-Dense	0.0490	0.0682	0.0391	0.0487	0.0783	0.1304
Turna-Dense	0.0446	0.0926	0.0420	0.0655	0.0522	0.1826
LaBSE-Dense	0.1036	0.1564	0.0942	0.1192	0.1304	0.2783
Turkish-e5-Large-Dense	0.1263	0.1897	0.1130	0.1422	0.1652	0.3478
LaBSE-Sparse-Dense	0.1830	0.2992	0.1623	0.2181	0.2435	0.5652
Turkish-e5-Large-Sparse-Dense	0.3018	0.3804	0.2739	0.3100	0.4522	0.6087

some input tokens with alternatives generated by a small generator network, then training a discriminative model to determine whether each token has been replaced by a generator sample. Since the performance of these models yields very low scores, we focus solely on the dense variation, rather than on sparse search models, which tend to produce poorer results. Additionally, we disregard their model size and runtimes for the same reason.

Turna [46] provides a Turkish language model designed to examine both natural language understanding and generation tasks. In this case, we use the Turna-Dense⁸ model, which is the largest model in our experimental setup, with a size of 4.6 GB. LaBSE⁹ is a language-agnostic BERT-based sentence embedding model that supports 109 languages [19]. The model size of LaBSE is 1.9 GB, which is almost half the size of the Turna model, yet it outperforms Turna in terms of R@10 and all other metrics.

The Turkish-e5-Large model¹⁰ has a size of 2.3 GB and is a fine-tuned version of the multilingual-e5-large-instruct model, trained on various Turkish datasets [29]. So far, we have collected sample texts and meanings for metaphorical expressions and have excluded sparse-based models such as BM25 [40] and SPARTA [56] from our analysis of these expressions. However, we now combine BM25 as the sparse model with the Turkish-e5-Large dense model. While the dense model focuses on textual representations, the sparse and dense variation incorporates both keywords and textual representations. In this context, the runtime for this combination per query is 0.11176 seconds for the candidate metaphor detection subtask.

In Table 2, M@3, M@10, R@3, and R@10 represent the MAP@3, MAP@10, Recall@3, and Recall@10 metrics, respectively. Experimental results in Table 2 show that the combination of sparse and dense variations of the Turkish-e5-Large model outperforms the dense-only variation of the same model. Furthermore, it surpasses the sparse and dense variation of the LaBSE model, which is why we continue to use the Turkish-e5-Large model for subsequent candidate metaphor identification tasks.

4.4 Comparison of Models for Candidate Metaphor Identification

As the second step, we evaluate several pre-trained Turkish NLI models, including the BERT base, DistilBERT base, and ConvBERT base models. The BERTNLI base model¹¹ is trained on Turkish machine-translated versions of the NLI and STS-B (Semantic Textual Similarity) datasets, which include text from image captions, news headlines, and user forums [10], using the Sentence Transformer training mechanism.

⁸https://huggingface.co/boun-tabı-LMG/turna_semantic_similarity_stsb_tr

⁹<https://huggingface.co/sentence-transformers/LaBSE>

¹⁰<https://huggingface.co/ytu-ce-cosmos/turkish-e5-large>

¹¹<https://huggingface.co/emrean/bert-base-turkish-cased-mean-nli-stsb-tr>

Table 3. Comparison of Candidate Metaphor Identification Models

Model	NDCG@3	NDCG@10	M@3	M@10	R@3	R@10
BertNLI-stanza	0.3388	0.4630	0.2942	0.3563	0.4348	0.7652
DistilbertNLI-stanza	0.3660	0.4967	0.3304	0.3962	0.4696	0.8174
ConvbertNLI-stanza	0.3400	0.4827	0.3000	0.3607	0.4435	0.8261
SimCSE-TR-stanza	0.4619	0.5733	0.4174	0.4756	0.5913	0.8783
SimCSE-TR-zemberek	0.4413	0.5581	0.3957	0.4559	0.5739	0.8783
SimCSE-TR-trmorph	0.4216	0.5335	0.3783	0.437	0.5478	0.8348

The DistilBERTNLI base model ¹² is a fine-tuned version of the Turkish DistilBERT base model, developed by the MDZ Digital Library team. The ConvBERTNLI base model ¹³ is another fine-tuned version of the DBMDZ ConvBERT base model for Turkish. Furthermore, we employ the SimCSE-TR model ¹⁴ as a pre-trained model based on Turkish Wikipedia and Turkish NLI. In its supervised form, the model uses annotated sentence pairs from the NLI dataset as the contrastive training objective, with entailment pairs treated as positive examples and contradiction pairs as hard negative examples [47].

In our work, we randomly select 400 target terms, each containing one metaphorical use in a sentence, along with its meaning as a positive example. We also randomly select one literal use in a sample with its meaning as a hard negative example for each target term. As part of the supervised contrastive setting, we explicitly select negative examples from the set of training sentences that are neither paraphrases nor part of the same context. In this case, negative pairs may come from any other random sentence in the training dataset that is dissimilar.

Table 3 compares the performance of these models for the top-k (where k represents at least two candidate expressions) in the metaphor identification task. In this scenario, the SimCSE-TR model outperforms the other models on all metrics, with each model generating stemmed tokens using the Stanza library for Turkish.

As part of the ablation study, we compare other Turkish libraries, such as Zemberek-NLP ¹⁵ and TRmorph [14], with the best-performing SimCSE-TR model. Experimental results demonstrate that both the Stanza and Zemberek-NLP models achieve the same R@10 score of 0.8783, which is superior to the TRmorph-based SimCSE-TR model. Considering other metrics, the Stanza-based model outperforms the Zemberek-NLP-based model.

4.5 Comparison of Reranking Scenarios

We compare different reranking scenarios using the meanings, sample texts, and combinations of both for the candidate expressions. In this comparison, Di-Sample and Di-Meaning represent the model-based embeddings of DistilBERTNLI with sample texts and meanings, respectively. Di-Contr-Sample is the fine-tuned model that leverages contrastive learning. Similarly, Co-Sample refers to the ConvBERTNLI model with sample embeddings, while Co-Meaning represents the meaning embeddings of the ConvBERTNLI model. Furthermore, Co-Contr-Sample is the fine-tuned version of the ConvBERTNLI model based on contrastive learning.

The SimCSE-TR-Contr-sample-meaning is the fine-tuned version of the SimCSE-TR model, using both sample texts and meanings of 400 target terms, with respect to positive and negative examples. All contrastive learning models follow the same supervised contrastive setting as previously

¹²https://huggingface.co/emrean/distilbert-base-turkish-cased-allnli_tr

¹³https://huggingface.co/emrean/convbert-base-turkish-mc4-cased-multinli_tr

¹⁴<https://huggingface.co/ardauzunoglu/sup-simcse-tr-bert-base>

¹⁵<https://github.com/ahmetaa/zemberek-nlp>

Table 4. Comparison of Experimental Results for Reranking Scenarios

Model	NDCG@3	NDCG@10	M@3	M@10	R@3	R@10
Di-Sample	0.3660	0.4967	0.3304	0.3962	0.4696	0.8174
Di-Meaning	0.2647	0.3558	0.2420	0.2878	0.3304	0.5739
Co-Sample	0.3400	0.4827	0.3000	0.3607	0.4435	0.8261
Co-Meaning	0.3345	0.4552	0.2087	0.2457	0.4348	0.7565
Di-Contr-Sample	0.3872	0.5184	0.3730	0.4286	0.4886	0.8381
Di-Contr-Meaning	0.2914	0.3744	0.2733	0.3244	0.3621	0.6028
Co-Contr-Sample	0.3596	0.4772	0.3404	0.3892	0.4589	0.8411
Co-Contr-Meaning	0.4085	0.4996	0.2309	0.2688	0.6174	0.7982
SimCSE-TR-Contr-meaning	0.4946	0.6085	0.4464	0.5055	0.6348	0.9304
SimCSE-TR-Contr-sample	0.5024	0.6229	0.4507	0.5131	0.6522	0.9652
SimCSE-TR-Contr-sample-meaning	0.5857	0.6871	0.5478	0.5971	0.6956	0.9739

discussed, explicitly selecting negative examples from the set of training sentences that are neither paraphrases nor part of the same context.

Table 4 shows that the contrastive learning models outperform all other models. Among these, SimCSE-TR-Contr-Sample-Meaning achieves the highest Recall@10 (R@10) of 0.9739 on the generated test dataset. Additionally, the runtime for the metaphor identification subtask using the SimCSE-TR-Stanza with this reranking model per query is 0.57434 seconds, and the model size is 885.3 MB for the SimCSE-TR model.

4.6 Error Analysis

Table 5 presents a detailed error analysis, focusing on the target word [acı], which can mean either ‘pain’ or ‘bitterness’ in Turkish. As Turkish is an agglutinative language, there are several derivational forms of the root word, such as [acılı] (spicy), or compound phrases like [acı tatlı] (bittersweet), which include multiple terms. These variations can introduce challenges in correctly identifying the metaphorical or literal use of the target word.

If the meaning of the target word, as shown in the first sample, clearly identifies the term, the model correctly predicts the target word as literal (L). For instance, the first meaning refers to the bitter taste of coffee, which is a literal usage. In contrast, the second sample text clearly demonstrates the metaphorical (M) usage of the same target term.

However, detecting metaphorical or literal usage becomes more challenging when the target word consists of two terms, as seen in the fourth sample [acı tatlı] (bittersweet). This phrase contains two distinct words, making it more difficult to differentiate between literal and metaphorical usage due to the limited context.

We also categorise the different error types that impact the performance of the metaphor detection and identification subtasks in Table 6. Since it employs a dictionary-based methodology, some target expressions may consist of one-word meanings, which, in particular, makes it unsuitable for dense search models. Furthermore, the lemmatiser can generate a different form due to the circumflex on certain characters, such as ‘â’. Regarding affix handling, the root form may have various representations, such as noun or verb forms. Another critical issue is the infinitive form of the verb stem, as the suffixes ‘-mek’ or ‘-mak’ can be influenced by either front or back vowels.

The error analysis highlights the need for improvements in the Turkish lemmatiser and handling of suffixes in target words as a focus for future updates. Another critical task will be enhancing the understanding of the target words’ meanings by incorporating relations among synsets, such as hypernymy, which connects more general meanings to their specific counterparts. Additionally, relational adjectives could be incorporated into knowledge graphs to represent antonyms. Direct antonyms reflect a profound semantic contrast between the respective concepts, while each of

Table 5. Error Analysis of Predicted and Actual Samples

Sample text	Meaning	Actual	Predicted
[Acı] kahvesini yudumluyordu. (She was sipping her bitter coffee.)	Tadı bu nitelikte olan (Having this quality of taste)	L	L
[Acı] poyraz kuvvetle esiyordu. (The bitter north wind was blowing strongly.)	Keskin, şiddetli (Sharp, severe)	M	M
[Acı su] sabunu köpürtmez. (Bitter water does not foam soap.)	İçindeki minerallerin etkisiyle tadı sert olan kuyu veya pınar suyu (Well or spring water with a strong taste due to the effect of the minerals it contains)	L	L
İşi bir ucundan tuttuk, [acı tatlı] konuşup durduk. (We took hold of the matter from one end, and kept talking in a bittersweet way)	Şöyle böyle (More or less)	L	M
[Acılı] tarhana. (Spicy tarhana.)	Acı katılmış olan (The bitterness is added)	L	L
Pek [acı] bir geçmişi bulunan roman kahramanım burada bütün dertlerinden kurtulur. (My novel hero, who has a very bitter past, will be freed from all of his troubles here)	Acısı olan, kederli (One who has pain, sorrowful.)	M	M
Henüz, yüreğimin bir yanı da söyleyememiş, dışı atılamamış [acılıklarla] dolu. (Yet, there is a part of my heart that is full of bitterness that cannot be said, that cannot be expressed.)	Dokunaklılık, kederlilik, yashılık (Touchiness, sorrowfulness, mournfulness)	M	M
General, soluk dudaklarını parmaklarının arasına alarak [acı acı] gülüyor. (The general, laughing bitterly with his pale lips held between his fingers)	Üzüntülü bir biçimde, dokunaklı olarak (In a sorrowful manner, poignantly)	M	L

Table 6. Categorisation of the Different Types of Errors Affecting Overall Performance

Category	Example
The meaning of a term might be expressed in very short texts, particularly a single word.	The word “şık” (option) has a meaning that is also a single word, namely “seçenek” (which also means “option” in English, though it is a different word in Turkish).
The lemmatiser can generate a different form and may fail to retrieve the correct lemma from the dictionary.	The case of “hâl” (situation), which is lemmatised as “hal” (without the circumflex).
Lemmatisation can result in different representations.	“aç” (hungry) represents a noun, while “aç-tı” (opened) indicates the verb form.
Due to affix handling, it is a challenging task to reach the most appropriate form.	“ağır” (heavy) can be transformed into “ağ-rı-mak” (to ache).
The infinitive form issue of the verb stem is influenced by whether the verb contains front vowels (e, i) or back vowels (a, ı).	“git-mek” (to go) and “aç-mak” (to open).

these polar adjectives is linked to a set of semantically similar adjectives. These similar adjectives, in turn, serve as indirect antonyms, linking to the opposing member’s meaning, enriching the semantic contrast.

5 Discussion on Real-World Use Cases

To demonstrate the generalisation capability of our method, we prepared a real-world dataset, including real news, social media, and literary texts, to serve as a measure of external validity, rather than relying on test data in dictionary format. Table 7 illustrates the performance of our method on this dataset, which consists of 114 target terms, each with at least one metaphorical and one literal sentence representation. For the metaphor detection subtask, we employ the Turkish-e5-Large

Table 7. Performance Evaluation on the Real-World Dataset

Task	NDCG@3	NDCG@10	M@3	M@10	R@3	R@10	Model size	Runtime in secs
Metaphor detection	0.3263	0.4013	0.2968	0.334	0.4123	0.614	2.3G	0.10114
Metaphor identification	0.6324	0.687	0.5994	0.628	0.7281	0.8684	885.3M	0.34911

model as a fine-tuned version of the multilingual-e5-large-instruct model [49]. During the training phase, the multilingual-e5 model undergoes weakly-supervised contrastive pre-training on billions of text pairs, followed by supervised fine-tuning on a small set of high-quality labelled datasets. For the metaphor identification subtask, we utilise the SimCSE-TR model [47], a pre-trained model based on Turkish Wikipedia and Turkish NLI.

The performance evaluation in Table 7 indicates that our metaphor detection subtask, using the Turkish-e5-Large model with sparse and dense search variations, achieves a 0.614 Recall@10 (R@10) score. This is slightly better than the performance of the same model (0.6087 R@10 score in Table 2), evaluated on the dictionary-based test dataset. Furthermore, the metaphor identification results show that, by using the same zero-shot SimCSE-TR model for both datasets, our method, which leverages sample and meaning representations, outperforms the same model on the real-world dataset across all metrics except R@10. In this case, it achieves a competitive R@10 score of 0.8684, compared to 0.9739 in Table 4. These results clearly demonstrate that our model can generalise to this real-world dataset. In terms of model size, the Turkish-e5-Large model requires 2.3 GB for the metaphor detection subtask. By contrast, the SimCSE-TR model occupies 885.3 MB for the metaphor identification subtask. When comparing runtimes for each example, the duration for the identification task is almost three times slower than for the detection task, taking 0.34911 seconds.

These promising experimental results show that our method can be adapted to identify and detect metaphorical language across several real-world applications. One prominent use case is metaphor-enabled market sentiment analysis [33], which aims to enhance the detection of sentiment-driven trends. By understanding metaphorical expressions in financial contexts, this approach allows for more accurate and effective communication of asset projections and market insights.

In software engineering, communication analysis is crucial, as developers often use metaphorical expressions to convey complex concepts in more intuitive terms. Imran et al. [25] examine the impact of such metaphorical language on fine-tuned language models by analysing GitHub comments, issues, and pull requests. Their research highlights how metaphors can shape the understanding and interpretation of technical discussions, potentially guiding code reviews and enhancing collaboration.

Another critical application is in the field of mental health, particularly in the study of symptom-metaphor interactions. In trauma evaluation and treatment, metaphorical expressions are often used to describe emotional and psychological states. Qiu et al. [39] suggest that recognising these metaphors can provide deeper insights into an individual's psychological condition. By understanding how metaphors reflect emotional and cognitive states, they aim to help mental health professionals interpret trauma symptoms, leading to more effective treatment plans and personalised support strategies.

Furthermore, in dialogue systems, such as conversational multi-agent systems, distinguishing between figurative and literal language is essential. Dialogue systems that recognise metaphorical expressions can enhance interactions by tailoring responses to the user's emotions. According to Dybala et al. [17], incorporating humour into conversational systems based on user emotions can improve the overall user experience. These systems determine when and how to use humour, which

can increase engagement and provide a more positive interaction compared to standard agents that lack such emotional awareness.

6 Conclusion and Future Work

In this study, we present a method for Turkish metaphor detection and identification in texts using a contrastive learning approach. In the metaphor detection subtask, performance evaluation shows that sparse and dense search variations using the Turkish-e5-Large model achieve a Recall@10 (R@10) score of 0.614. Furthermore, the SimCSE-TR-Contr-Sample-Meaning model achieves the highest R@10 score of 0.9739 on the generated test dataset for the metaphor identification subtask. In a real-world scenario, it achieves a competitive R@10 score of 0.8684, demonstrating that our model can generalise effectively to such contexts.

Future work will focus on integrating knowledge-based approaches into the proposed method. We also plan to explore semantic path similarity techniques using knowledge graphs to further enhance the performance of sentence embedding models. Another direction for future research will involve adapting our approach to detect irony in Turkish texts [6].

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Appendices

A Metaphor Dataset Generation Algorithm

To generate a metaphor identification and detection dataset, we first sort the original dataset (excluding examples used in the fine-tuning task) in descending order to identify the most common metaphorical and literal expressions in the Turkish dictionary. We then filter the dataset based on these terms, ensuring that each term has at least one metaphorical and one literal usage. Additionally, we only collect samples where the total number of metaphorical and literal representations is greater than or equal to three.

As indicated in the generated dataset, the number of literal samples tends to exceed the number of metaphorical samples. However, by employing the retrieval-based methodology, the issue of unbalanced datasets is addressed through the first dense retrieval-based search mechanism, which retrieves highly relevant candidates for the given input and target term. Without fine-tuning, we compare the performance of Turkish pre-trained models.

The algorithm for generating the training and test datasets consists of two main steps. First, it generates a term list from the Turkish Wiktionary by retrieving terms from the Turkish WordNet and matching them to Wiktionary entries using exact string matching. Additionally, the Jaccard similarity between definitions in WordNet and the Wiktionary is calculated to refine the term list. The second step involves extracting detailed information from the Turkish dictionary, where we collect specific details for each term, including the target entry, meaning, classification label (metaphorical or literal), and sample sentences.

In this section, we present the algorithm for generating training and test datasets for metaphor identification and detection, which is divided into three sub-algorithms. Algorithm 1 extracts the definition for each term t from the Turkish Wiktionary, which contains terms ordered alphabetically from A to Z, and generates the term list L_t . Algorithm 2 outlines the process of retrieving terms from the Turkish WordNet [18] and matching them to entries in the Turkish Wiktionary using the exact string matching algorithm (`ComputeExactStringMatching()`). Additionally, it computes the Jaccard similarity (`ComputeJaccardSimilarity()`) between the definitions of the terms in WordNet and the Turkish Wiktionary, resulting in a more comprehensive term list, L_t . Algorithm 3 focuses on the Turkish dictionary, denoted as D , where we extract detailed information for a given term D_t if it exists in the dictionary. This detailed information includes the target entry e , the meaning m , the property p (which can be either metaphorical or literal), and the sample text s associated with the term.

ALGORITHM 1: Term extraction from the Turkish Wiktionary.**Input:** URL of the Turkish Wiktionary page**Output:** List of extracted terms L_t

```

1 Function ExtractTermsFromWiktionary():
2    $L_t \leftarrow$  empty list;
3   doc  $\leftarrow$  connect to the URL using Jsoup;
4   termLinks  $\leftarrow$  select all links from div id where a[href id];
5   for each link in termLinks do
6      $t \leftarrow$  text of the link;
7      $L_t \leftarrow$  add( $t$ );
8     nextPageLink  $\leftarrow$  select the link for the next page if it exists;
9     if nextPageLink is not null then
10      nextPageUrl  $\leftarrow$  concatenate base URL with nextPageLink.href;
11       $L_t \leftarrow$  addAll(ExtractTermsFromNextPage(nextPageUrl));
12    end
13  end
14  return  $L_t$ ;

```

ALGORITHM 2: Exact string matching and Jaccard similarity calculation for the term list.**Input:** Term list $\{t_1, t_2, \dots, t_n\}$ extracted from the Turkish Wiktionary**Output:** Comprehensive term list L_t with similarity scores

```

1 Function ComputeExactStringMatching(terms, definitions):
2   exactMatches  $\leftarrow$  empty list;
3   for each pair of terms ( $t_i, t_j$ ) in terms do
4     definition_i  $\leftarrow$  definition of term  $t_i$ ;
5     definition_j  $\leftarrow$  definition of term  $t_j$ ;
6     if definition_i == definition_j then
7       exactMatches.add( $t_i, t_j$ );
8     end
9   end
10  return exactMatches;
11 Function ComputeJaccardSimilarity(terms, definitions):
12  jaccardScores  $\leftarrow$  empty list;
13  for each pair of terms ( $t_i, t_j$ ) in terms do
14    definition_i  $\leftarrow$  definition of term  $t_i$ ;
15    definition_j  $\leftarrow$  definition of term  $t_j$ ;
16    set_i  $\leftarrow$  set of unique words in definition_i;
17    set_j  $\leftarrow$  set of unique words in definition_j;
18    intersection  $\leftarrow$  set_i  $\cap$  set_j;
19    union  $\leftarrow$  set_i  $\cup$  set_j;
20    jaccardScore  $\leftarrow \frac{|intersection|}{|union|}$ ;
21    jaccardScores.add( $t_i, t_j$ , jaccardScore);
22  end
23  return jaccardScores;

```

ALGORITHM 3: Extraction of detailed information from the Turkish dictionary.

Input: W Turkish Wiktionary, Wo Turkish WordNet, t term, L_t term list, D Turkish dictionary, e target entry, m meaning, p property (either metaphor or literal), s sample text.

Output: L_e entry list including the detailed information for each term D_t .

```

1  $WConnection \leftarrow connectWiktionary(W)$ ;
2 if  $WConnection.IsSuccessStatusCode()$  then
3    $L_t \leftarrow ExtractTermsFromWiktionary$ ;
4    $L_t \leftarrow ComputeExactStringMatching$ ;
5    $L_t \leftarrow ComputeJaccardSimilarity$ ;
6   foreach  $t$  in  $L_t$  do
7      $DConnection \leftarrow connectDictionary(D)$ ;
8     if  $DConnection.isExist(D_t)$  then
9        $e \leftarrow DConnection.query(D_t)$ ;
10       $m \leftarrow DConnection.query(D_t)$ ;
11       $p \leftarrow DConnection.query(D_t)$ ;
12       $s \leftarrow DConnection.query(D_t)$ ;
13       $L_e \leftarrow addInformation(e, m, p, s)$ ;
14   end
15 end
16 end

```

B Comparison with Literature Studies

We evaluate our method against two state-of-the-art studies, MelBERT and MisNet, both of which are publicly available and share similar objectives. MelBERT [11] employs a Siamese architecture with BERT and RoBERTa models, encoding both a sentence containing a target word and the target word itself to capture contextualised and isolated meanings. Despite reducing computational cost through a metaphor-aware late interaction approach, it requires a large English dataset (VUA-20 [32]), which is unbalanced, with only 12% of sentences containing metaphorical expressions. MisNet, also based on a Siamese network, combines semantic matching of the target word's contextual and basic meanings with interaction-based semantic mismatch. It also relies on a large dataset (VUA-All [32]) and external resources, such as the Lexico dictionary. In contrast, our method avoids a dedicated training phase and does not depend on well-formed or balanced datasets. To demonstrate the advantage of our approach, we train both MelBERT and MisNet using only 50 and 500 examples from the Turkish dataset, ensuring a fair comparison by using the Turkish RoBERTa-small model [3] in a GPU-limited environment.

Figure 2 compares the proposed method with MelBERT and MisNet, showing that our approach, which fine-tunes a pre-trained Turkish NLI model (Co-Contr-Sample in Table 4) without a training phase, outperforms both models even with limited training data. Our method achieves a 0.557 F1 score when the training size is limited to just 50 metaphorical and 50 literal examples. Even without any training samples, our approach still outperforms both models, achieving a 0.518 F1 score. As the training dataset size increases to 500 examples, MisNet performs best with an F1 of 0.801, while our method reaches an F1 score of 0.733, demonstrating its competitiveness when compared to supervised learning methods. Unlike MisNet and MelBERT, which focus on word classification and sequence labelling, respectively, our method detects metaphorical usage without directly defining the target word. Additionally, it leverages a comprehensive Turkish dictionary, enhancing its performance and maintaining competitiveness in zero-shot settings.

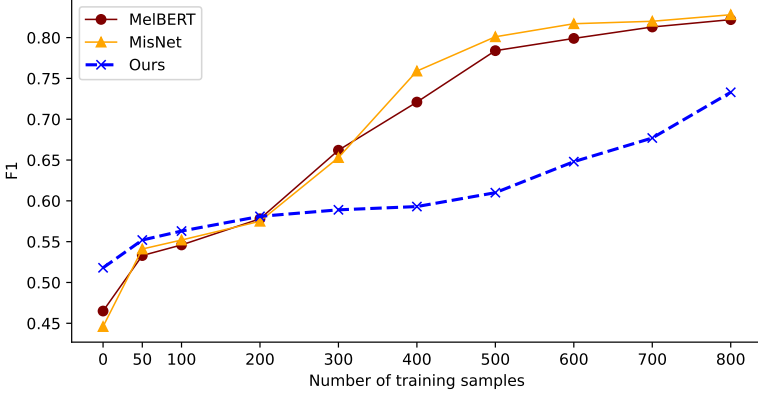


Fig. 2. Comparison of our model with others using different training sample sizes.

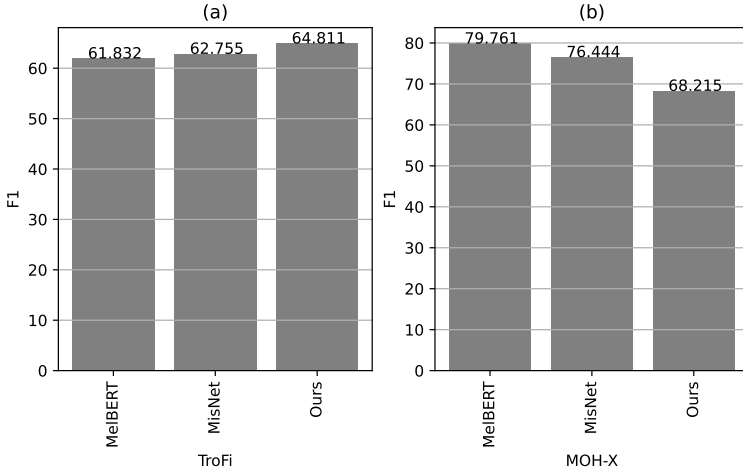


Fig. 3. Comparison of preliminary results for our method on the Trofi and MOH-X datasets in English.

Figure 3 illustrates a setup similar to that in Figure 2, where only 50 training samples are used for contrastive learning with the SNLI dataset [7]. Our approach follows the BEIR framework for DPR-based semantic search to identify metaphorical expressions, using English lemmatisation through the Stanza library to query a Vector DB Semantic dictionary curated from metaphorical expressions and their meanings.

The results show that our method outperforms both MelBERT and MisNet on the Trofi dataset, achieving an F1 score of 64.811. On the MOH-X dataset, while MelBERT achieves the highest F1 score of 79.761, our method remains competitive with an F1 score of 68.215.

Transformers, such as the BERT model, use a scaled dot-product attention mechanism [48], with a complexity of $\mathcal{O}(n^2 \cdot d)$, where n is the sequence length and d is the dimension of the vector representation. Multi-head attention increases this complexity to $\mathcal{O}(n^2 \cdot d + n \cdot d^2)$, as it applies different linear transformations per head to capture distinct relationships. The RoBERTa model, which serves as the backbone for both the MelBERT and MisNet models, improves upon BERT

by using dynamic masking, increasing batch size, and removing next-sentence prediction loss. Both models employ two RoBERTa base models as encoders for sentences and target words. In contrast, our method fine-tunes the Turkish ConvbertNLI model, which uses span-based dynamic convolution to capture local token dependencies with linear complexity, making it efficient for distinguishing meanings. The SimCSE-based contrastive learning mechanism proves effective with scarce annotated data, using dropout as an augmentation method. A recent study [2] investigates the optimal sample complexity for contrastive learning and shows that the sample complexity under cosine similarity has an upper bound of $\mathcal{O}(n \cdot \min(n, d))$. Finally, our method is faster, with training epochs taking around 4 minutes, compared to MelBERT and MisNet, which are more than twice as slow.

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